CSA1713 – Artificial Intelligence

Name:- B. Deepthi reddy

Reg No:- 192312288

Slot – C

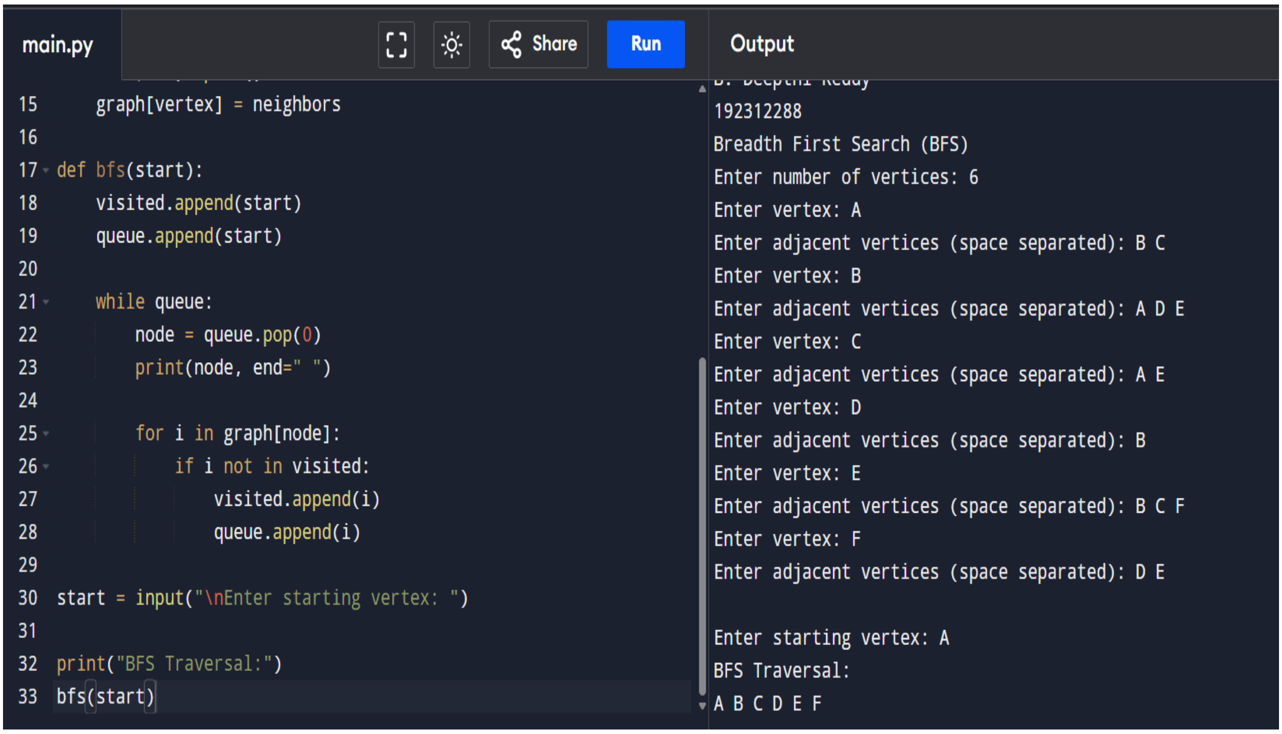
5. BREADTH FIRST SEARCH (BFS)

Code:

```
print("B. Deepthi Reddy")
print(192312288)
print("Breadth First Search(BFS)")
visited = []
queue = []

graph = {}

n = int(input("Enter number of vertices: "))

for i in range(n):
    vertex = input("Enter vertex: ")
neighbors = input("Enter adjacent vertices (space separated): ").split()
    graph[vertex] = neighbors

def bfs(start):
```

```
visited.append(start)
queue.append(start)

    while queue:
        node = queue.pop(0)
print(node, end=" ")
        for i in graph[node]:
            if i not in visited:
visited.append(i)
queue.append(i)
start = input("\nEnter starting vertex: ")
print("BFS Traversal:")
bfs(start)
```

Output:



main.py

```python
print("B. Deepthi Reddy")
print(192312288)
print("Vacuum Cleaner Problem")

A = input("Enter Room A status (Dirty/Clean): ")
B = input("Enter Room B status (Dirty/Clean): ")

print("\nInitial State")
print("Room A =", A)
print("Room B =", B)

print("\nVacuum starts at Room A")

if A.lower() == "dirty":
    print("Room A is Dirty")
    print("Cleaning Room A...")
    A = "Clean"
else:
    print("Room A is already Clean")
```

Output

```
B. Deepthi Reddy
192312288
Vacuum Cleaner Problem
Enter Room A status (Dirty/Clean): Clean
Enter Room B status (Dirty/Clean): Dirty

Initial State
Room A = Clean
Room B = Dirty

Vacuum starts at Room A
Room A is already Clean

Move to Room B
Room B is Dirty
Cleaning Room B...

Final State
Room A = Clean
```